Enhancement Plan for eQual Framework Journal Paper

A Hybrid Evaluation Framework for Gender-Biased Job Ads with Natural Language Processing

Date: October 15, 2026
Purpose: Comprehensive enhancement plan to address reviewer comments and reposition the paper as a framework rather than a complete system

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EXECUTIVE SUMMARY

This enhancement plan addresses critical reviewer feedback and repositions the eQual system from a complete implementation to a comprehensive NLP-based evaluation framework for gender-biased job postings. The plan integrates insights from 696 related studies across five literature domains to strengthen methodological rigor, validation mechanisms, and generalization capabilities.

Key Strategic Shifts:

1. From System to Framework: Emphasize the modular, extensible architecture that can be adapted to different contexts
2. Strengthen Validation: Add cross-domain testing, external validation, and fairness-aware evaluation metrics
3. Address Data Scarcity: Implement data augmentation and class imbalance mitigation strategies for implicit bias
4. Enhance Reproducibility: Provide detailed implementation specifications and hyperparameter configurations
5. Deepen Discussion: Connect findings to broader NLP fairness literature and practical ICT applications

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I. CRITICAL REVIEWER COMMENTS SUMMARY

Associate Editor & Layout Editor Requirements

- Structure: Mandatory IMRADC format (Introduction, Method, Results and Discussion, Conclusion)
- Method Section: Must be detailed enough for replication with logical step arrangement
- Discussion Section: 6-step structure (gaps, key findings, comparison, limitations, future work, conclusion)
- Citations: Sequential IEEE format, minimum 25 references (15 recent journal articles)
- Formatting: Complete author biographies with profile links (Scholar, Scopus, Publons, ORCID)

Reviewer A - Methodological Concerns

1. Methodology too long and proposal-like: Distinguish actual implementation from conceptual design

2. Results lack focus: Separate core empirical results from additional results
3. Missing flowchart: Add methodology flowchart for research pipeline clarity
4. Citation issues: Ensure all references are cited in text, fix sequential numbering
5. Table integration: Introduce each table in text before display

Reviewer B - High-Impact Recommendations

1. Complete human evaluation: Finalize and report human evaluation for rewrite module (neutrality, semantic preservation, clarity)
2. Address data imbalance: Provide deeper analysis/mitigation for limited implicit bias samples
3. Strengthen validation: Add external datasets or cross-domain testing to prevent overfitting
4. Improve reproducibility: Include detailed implementation (model configurations, training parameters, tools, hardware)
5. Enhance ICT relevance: Explain real-world application in recruitment and public employment platforms

Reviewer C - Technical Consistency

1. Table references: Explicitly reference each table in narrative before presentation
2. Citation irregularity: Fix inconsistent reference numbering sequence
3. Template compliance: Ensure adherence to IJICT journal standards

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II. CURRENT PAPER STRUCTURE & GAPS

Current System: eQual

7-Phase Architecture:

1. Data Collection (1,501 PESO/DOLE job ads from Facebook)
2. Annotation (Gold-400 + Pilot-30 subsets)
3. Bias Detection (Rule-based, TF-IDF, Transformer models)
4. Span Localization (Identifying biased text segments)
5. Neutral Rewriting (Generating inclusive alternatives)
6. Fairness-Aware Recommendation (Ranking neutral alternatives)
7. Usability Study (SUS evaluation)

Identified Gaps:

- Implicit bias samples: Very limited (dataset shows empty "Gender-Biased Words/Phrase/Statement" columns)
- Human evaluation: Incomplete for rewrite module
- External validation: No cross-domain or external dataset testing
- Implementation details: Insufficient for full replication
- Generalization: Potential overfitting concerns not addressed
- Framework positioning: Presented as complete system rather than adaptable framework

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III. LITERATURE-INFORMED ENHANCEMENT STRATEGY

A. Methodologies for Gender-Bias Detection (Explicit & Implicit)

From Literature Review (91 papers):

1. Explicit Bias Detection

Current Approach: Rule-based lexicon matching
Enhancement Needed: Context-aware discrimination detection

Key Insights:

- Context-aware models (RoBERTa, BERT) significantly improve precision over keyword matching 1
- Simple term filtering (e.g., "young company" vs. "young candidates") causes false positives 1
- Transformer-based models achieve 95%+ accuracy in contextual bias detection 2 , 5

Recommended Enhancement:

Phase 3a: Explicit Bias Detection
- Baseline: Rule-based lexicon (current)
- Enhanced: RoBERTa-based contextual classifier
- Validation: Compare precision/recall on Gold-400 subset
- Metrics: Precision, Recall, F1-score, False Positive Rate

2. Implicit Bias Detection

Current Challenge: Scarcity of implicit bias examples in dataset
Critical Gap: No mitigation strategy for class imbalance

Key Insights from Literature:

- Implicit bias includes gender-coded traits (agentic vs. communal language) 3 , 6 , 7
- Contrastive learning improves detection of subtle biases 8
- Masked Language Models (MLMs) can quantify implicit associations 8 , 13
- Three-category framework: agentic (masculine-coded), balanced, communal (feminine-coded) 3

Recommended Enhancement:

Phase 3b: Implicit Bias Detection Framework
- Taxonomy: Adopt agentic-balanced-communal classification [3]
- Model: Fine-tuned RoBERTa with contrastive learning
- Data Augmentation: Address class imbalance (see Section III.E)
- Validation: Template-based probing + counterfactual testing

3. Unified Detection Architecture

Framework Recommendation:

Multi-Level Bias Detection Module:
├─── Level 1: Explicit Bias (Rule-based + RoBERTa)
├─── Level 2: Implicit Bias (Agentic-Communal Classifier)
├─── Level 3: Intersectional Bias (Gender × Domain)
└─── Output: Bias scores + localized spans + category labels

B. Neutral Rewriting with Semantic Preservation

From Literature Review (196 papers):

1. State-of-the-Art Approaches

Key Insights:

- Gradient-based rewriting (DEPEN): Detects sensitive components, masks them, perturbs generation under neutralizing constraint 20
- Agency-power counterfactuals: Jointly generates alternatives that balance agency and power 21
- Gender-neutral rewriting models: NeuTral Rewriter achieves <0.18% WER on synthetic/in-domain/out-domain tests 27
- Controlled paraphrase generation: Syntax-guided approaches preserve semantic meaning 28
- LLM-based rewriting: ChatGPT/GPT-4 can generate gender-neutral alternatives 2 , 19

Current Gap: No reported human evaluation for neutrality, semantic preservation, clarity

Recommended Enhancement:

Phase 5: Enhanced Neutral Rewriting Module
├─── Approach 1: Rule-based substitution (baseline)
├─── Approach 2: Fine-tuned T5/BART for controlled rewriting
├─── Approach 3: LLM-guided rewriting (GPT-4 with prompts)
└─── Evaluation Framework:
  ├─── Automatic Metrics: BLEU, ROUGE, BERTScore, WER
  ├─── Semantic Preservation: Sentence-BERT similarity
  ├─── Bias Reduction: Before/after bias scores
  └─── Human Evaluation (MANDATORY):
    ├─── Neutrality (5-point Likert scale)
    ├─── Semantic Preservation (5-point Likert scale)

├─── Fluency/Clarity (5-point Likert scale)
├─── Acceptability (binary: acceptable/not)
└─── Annotators: N=3 per sample, minimum 100 samples

2. Semantic Preservation Techniques

- Sentence-BERT embeddings: Measure cosine similarity between original and rewritten text 20
- Controlled generation: Use syntax templates to maintain structure 28
- Iterative refinement: Multi-stage rewriting with validation 29

C. Validation Mechanisms for Classification & Recommendation

From Literature Review (80 papers):

1. Fairness Metrics for Classification

Key Insights:

- Prediction sensitivity: Measures fairness via counterfactual testing [Measuring Fairness 2022]
- Demographic Parity: P(Ŷ=1|A=a) should be equal across protected groups
- Equalized Odds: TPR and FPR should be equal across groups
- Counterfactual Fairness: Predictions should be invariant to protected attribute changes

Recommended Enhancement:

Validation Layer 1: Classification Fairness
├─── Intrinsic Metrics:
│ ├─── Demographic Parity Difference (DPD)
│ ├─── Equalized Odds Difference (EOD)
│ └─── Accuracy Parity
├─── Extrinsic Metrics:
│ ├─── Counterfactual Fairness Testing
│ ├─── Template-based Probing
│ └─── Stereotype Score Measurement
└─── Evaluation Protocol:
├─── Test on Gold-400 with gender-swapped pairs
├─── Measure performance disparity across groups
└─── Report fairness-accuracy tradeoffs

2. Recommendation System Validation

Key Insights:

- Fair ranking metrics: Exposure fairness, attention-weighted ranking fairness 17
- Multi-stakeholder fairness: Balance reader and author satisfaction [Fairness for both readers and authors 2023]
- Diversity metrics: Intra-list diversity, coverage of protected groups

Recommended Enhancement:

Validation Layer 2: Recommendation Fairness
├─── Ranking Fairness:
│ ├─── Normalized Discounted Cumulative Gain (NDCG)
│ ├─── Exposure Fairness (EF)
│ └─── Attention-Weighted Ranking Fairness (AWRF)
├─── Diversity Metrics:
│ ├─── Intra-List Diversity (ILD)
│ ├─── Coverage of neutral alternatives
│ └─── Semantic diversity of recommendations
└─── User Study:
├─── Perceived fairness (5-point scale)
├─── Usefulness of recommendations
└─── Acceptance rate of neutral alternatives

3. Comprehensive Evaluation Framework

Recommended Structure:

Multi-Dimensional Evaluation Framework:
├─── Model Performance:
│ ├─── Accuracy, Precision, Recall, F1-score
│ ├─── Confusion matrices
│ └─── Per-class performance breakdown
├─── Fairness Assessment:
│ ├─── Demographic Parity Difference
│ ├─── Equalized Odds Difference
│ ├─── Counterfactual fairness scores
│ └─── Bias amplification metrics
├─── Robustness Testing:
│ ├─── Adversarial examples
│ ├─── Perturbation analysis
│ └─── Stress testing with edge cases
├─── Human Evaluation:
│ ├─── Rewrite quality assessment
│ ├─── Usability study (SUS - already done)

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│   └── Expert review (HR professionals)
└── Computational Efficiency:
    ├── Inference time
    ├── Memory footprint
    └── Scalability analysis
```

D. Cross-Domain Testing & Generalization

From Literature Review (252 papers):

1. Domain Shift Challenges

Key Insights:

- Topic, domain, language shifts: Models degrade when test data differs from training [Bridging Topic, Domain, and Language Shifts 2023]
- Prompt-based learning: Improves domain generalization by unifying classification across domains [Prompt-based distribution alignment 2022]
- Transfer learning limitations: Fine-tuning can cause overfitting on source domain [SEDGE 2022]

Current Gap: No external validation or cross-domain testing reported

Recommended Enhancement:

```
Phase 8: Cross-Domain Validation (NEW)
├── Internal Validation:
│   ├── K-fold cross-validation on PESO dataset
│   ├── Temporal split (early vs. recent job ads)
│   └── Domain split (industry sectors)
├── External Validation:
│   ├── Dataset 1: International job ads (e.g., Australian job ads [2])
│   ├── Dataset 2: Academic job postings (STEM focus [3])
│   ├── Dataset 3: Private sector job ads (non-PESO)
│   └── Evaluation: Zero-shot and few-shot transfer
├── Generalization Metrics:
│   ├── Performance drop (source → target domain)
│   ├── Domain adaptation effectiveness
│   └── Robustness to distribution shift
└── Overfitting Detection:
    ├── Learning curves (train vs. validation loss)
    ├── Performance gap analysis
    └── Regularization effectiveness
```

2. Domain Adaptation Strategies

If performance degrades significantly:

Mitigation Strategies:
├─── Domain-Invariant Features:
│ ├─── Adversarial training for domain adaptation
│ └─── Contrastive learning across domains
├─── Few-Shot Adaptation:
│ ├─── Meta-learning for quick adaptation
│ └─── Prompt-based fine-tuning
└─── Ensemble Methods:
├─── Multi-domain model ensemble
└─── Domain-specific expert models

E. Mitigating Implicit Bias Data Scarcity

From Literature Review (75 papers):

1. Data Augmentation Techniques

Key Insights:

- LLM-based augmentation: GPT-4 generates high-quality synthetic samples 23 , [AugGPT 2023]
- Back-translation: Multi-level translation creates diverse paraphrases [MTDOT 2022]
- Contrastive sample construction: Generates hard negatives for better learning [Solving Data Imbalance 2023]
- Curriculum data augmentation: Progressively introduces augmented data during training [Few-Shot Text Classification 2021]

Current Challenge: Very limited implicit bias examples in PESO dataset

Recommended Enhancement:

Phase 2b: Data Augmentation for Implicit Bias (NEW)
├─── Strategy 1: LLM-Based Generation
│ ├─── Tool: GPT-4 with carefully designed prompts
│ ├─── Process:
│ │ ├─── Identify implicit bias patterns from literature
│ │ ├─── Generate job ad segments with implicit bias
│ │ ├─── Generate neutral alternatives
│ │ └─── Expert validation (HR professionals + linguists)
│ └─── Target: 500+ implicit bias examples

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├─── Strategy 2: Paraphrasing & Back-Translation
│    ├─── Paraphrase existing implicit bias examples
│    ├─── Back-translate through multiple languages
│    └─── Filter for quality and diversity
├─── Strategy 3: Counterfactual Data Augmentation
│    ├─── Gender-swap existing examples
│    ├─── Agentic ↔ Communal trait substitution
│    └─── Validate label consistency
└─── Quality Control:
     ├─── Semantic similarity check (threshold: 0.7-0.9)
     ├─── Toxicity screening
     ├─── Human spot-checks (N=100 samples)
     └─── Attribute verification (bias category preserved)
```

2. Class Imbalance Mitigation

Techniques from Literature:

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Imbalance Handling Framework:
├─── Data-Level:
│    ├─── Synthetic oversampling (SMOTE variants)
│    ├─── Borderline-SMOTE for boundary examples
│    └─── LLM-guided augmentation (preferred)
├─── Algorithm-Level:
│    ├─── Focal Loss (emphasizes hard examples)
│    ├─── Class-weighted loss functions
│    └─── Cost-sensitive learning
├─── Ensemble-Level:
│    ├─── Balanced bagging
│    ├─── EasyEnsemble
│    └─── RUSBoost
└─── Evaluation:
     ├─── Macro-averaged F1 (not micro)
     ├─── Per-class precision/recall
     ├─── Confusion matrix analysis
     └─── Minority class performance focus
```

3. Validation of Augmented Data

Quality Assurance Protocol:

Augmentation Validation:
├─── Automatic Checks:
│ ├─── Semantic similarity: 0.7 ≤ cosine_sim ≤ 0.9
│ ├─── Label consistency: Bias category preserved
│ ├─── Diversity: BLEU score < 0.8 (avoid near-duplicates)
│ └─── Fluency: Perplexity score within acceptable range
├─── Human Validation:
│ ├─── Sample: 10% of augmented data
│ ├─── Annotators: 3 experts (HR + linguistics)
│ ├─── Criteria: Realism, bias presence, label correctness
│ └─── Agreement: Krippendorff's α ≥ 0.7
└─── Ablation Study:
├─── Train with/without augmented data
├─── Compare performance on held-out test set
└─── Analyze impact on minority class performance

IV. REVISED PAPER STRUCTURE (IMRADC FORMAT)

1. INTRODUCTION

Current: Good foundation
Enhancements:

- Add paragraph on framework vs. system distinction
- Emphasize modularity and extensibility
- Clarify contributions more explicitly

Revised Contribution Statement:

This study proposes eQual, a modular NLP-based evaluation framework for gender-biased job postings in Philippine public employment contexts. The framework's key contributions include:

1. A comprehensive multi-level bias detection architecture that identifies both explicit and implicit gender bias using rule-based, statistical, and transformer-based approaches

2. A validated neutral rewriting module with human evaluation demonstrating semantic preservation and bias reduction

3. A fairness-aware recommendation system that ranks neutral alternatives while maintaining semantic fidelity

4. Extensive validation through cross-domain testing, fairness metrics, and usability studies (SUS score: [X])

5. Data augmentation strategies to address implicit bias scarcity, with quality-controlled synthetic data generation

6. A reproducible framework with detailed implementation specifications, enabling adaptation to other languages and employment contexts

2. RELATED WORK (NEW SECTION - Optional but Recommended)

Purpose: Position eQual within broader NLP fairness literature

Subsections:

- 2.1 Gender Bias in Job Advertisements
- 2.2 NLP Approaches to Bias Detection
- 2.3 Text Rewriting for Bias Mitigation
- 2.4 Fairness in Recommendation Systems
- 2.5 Research Gaps and Positioning

3. THE PROPOSED FRAMEWORK

Current Title: "Research Method"
Revised Title: "The Proposed eQual Framework" or "Framework Architecture and Methodology"

Enhancements:

- Add methodology flowchart (Reviewer A requirement)
- Clearly distinguish implemented components from conceptual design
- Simplify narrative, reduce technical jargon
- Organize by framework modules, not just phases

Revised Structure:

3. THE PROPOSED EQUAL FRAMEWORK

3.1 Framework Overview
- Architecture diagram (flowchart)
- Module descriptions
- Data flow
- Extensibility points

3.2 Data Collection and Annotation
- PESO/DOLE dataset (1,501 job ads)
- Annotation protocol
- Gold-400 and Pilot-30 subsets
- Inter-annotator agreement
[IMPLEMENTATION STATUS: COMPLETED]

3.3 Multi-Level Bias Detection Module
3.3.1 Explicit Bias Detection

- Rule-based lexicon approach
- RoBERTa contextual classifier
[IMPLEMENTATION STATUS: COMPLETED]

3.3.2 Implicit Bias Detection
- Agentic-communal classification
- Contrastive learning approach
[IMPLEMENTATION STATUS: COMPLETED/ENHANCED]

3.3.3 Span Localization
- Attention-based span extraction
[IMPLEMENTATION STATUS: COMPLETED]

3.4 Data Augmentation for Implicit Bias (NEW)
- LLM-based synthetic generation
- Quality control protocol
- Validation results
[IMPLEMENTATION STATUS: COMPLETED]

3.5 Neutral Rewriting Module
- Rewriting strategies (rule-based, neural, LLM-guided)
- Semantic preservation mechanisms
- Automatic evaluation metrics
[IMPLEMENTATION STATUS: COMPLETED]

3.6 Fairness-Aware Recommendation System
- Ranking algorithm
- Fairness constraints
- Diversity optimization
[IMPLEMENTATION STATUS: COMPLETED]

3.7 Validation and Evaluation Framework
3.7.1 Classification Validation
- Performance metrics
- Fairness metrics (DPD, EOD)
- Counterfactual testing

3.7.2 Rewriting Validation
- Automatic metrics (BLEU, BERTScore)
- Human evaluation protocol
- Results summary

3.7.3 Cross-Domain Validation (NEW)
- External datasets
- Generalization analysis
- Overfitting detection

3.7.4 Usability Study
- SUS evaluation

- User feedback
[IMPLEMENTATION STATUS: COMPLETED/ENHANCED]

3.8 Implementation Details
- Model configurations
- Hyperparameters
- Training procedures
- Hardware specifications
- Software dependencies
[IMPLEMENTATION STATUS: TO BE ADDED]

4. RESULTS AND DISCUSSION

Current: Data-rich but lacks focus
Enhancements:

- Separate core results from supplementary results
- Follow 6-step discussion structure (Associate Editor requirement)
- Add comparison with state-of-the-art methods
- Discuss limitations explicitly

Revised Structure:

4. RESULTS AND DISCUSSION

4.1 Introduction: Research Gaps Addressed
- Lack of end-to-end frameworks for Philippine context
- Limited implicit bias detection in job ads
- Absence of validated rewriting with semantic preservation
- Need for fairness-aware recommendation

4.2 Core Empirical Results
4.2.1 Bias Detection Performance
- Explicit bias: Precision, Recall, F1
- Implicit bias: Performance on augmented dataset
- Comparison: Rule-based vs. TF-IDF vs. Transformers
- Table X: Classification results on Gold-400

4.2.2 Neutral Rewriting Quality
- Automatic metrics: BLEU, ROUGE, BERTScore
- Semantic preservation: Sentence-BERT similarity
- Bias reduction: Before/after bias scores
- Table Y: Rewriting evaluation results
- **Human Evaluation Results (NEW):**
  - Neutrality: Mean=X.X, SD=X.X
  - Semantic Preservation: Mean=X.X, SD=X.X
  - Fluency: Mean=X.X, SD=X.X
  - Inter-rater reliability: Krippendorff's α=X.X

4.2.3 Recommendation System Performance

- Ranking quality: NDCG, MRR
- Fairness metrics: Exposure fairness
- User acceptance: Percentage of accepted recommendations
- Table Z: Recommendation evaluation results

4.2.4 Cross-Domain Validation (NEW)
- Performance on external datasets
- Generalization analysis
- Domain adaptation effectiveness
- Table W: Cross-domain performance comparison

4.3 Comparison with State-of-the-Art
- Comparison with [2] (Australian job ads)
- Comparison with [4], [6] (bias mitigation methods)
- Advantages of eQual framework
- Table: Comparative performance analysis

4.4 Fairness and Bias Analysis
- Demographic Parity Difference: X.XX
- Equalized Odds Difference: X.XX
- Counterfactual fairness scores
- Discussion of fairness-accuracy tradeoffs

4.5 Addressing Limitations
4.5.1 Data Limitations
- Implicit bias scarcity (mitigated via augmentation)
- Dataset size (1,501 ads)
- Domain specificity (PESO/DOLE context)

4.5.2 Model Limitations
- Potential overfitting (addressed via cross-domain testing)
- Computational requirements
- Language specificity (English/Filipino)

4.5.3 Evaluation Limitations
- Human evaluation sample size
- Annotator expertise
- Generalization to other contexts

4.6 Implications for Future Research
- Extension to other languages (e.g., Filipino, Tagalog)
- Integration with recruitment platforms
- Real-time bias detection and rewriting
- Multi-modal bias detection (text + images)
- Longitudinal studies on applicant diversity

4.7 Practical ICT Relevance (Reviewer B requirement)
- Integration with PESO/DOLE platforms
- Real-world deployment considerations

- Scalability and efficiency
- User training and adoption
- Policy implications for inclusive recruitment

4.8 Conclusion of Results
- Summary of key findings
- Framework effectiveness demonstrated
- Contributions to NLP fairness research
- Practical value for inclusive recruitment

5. CONCLUSION

Current: Adequate
Enhancements:

- Emphasize framework contributions
- Highlight validation rigor
- Discuss broader impact

Revised Structure:

5. CONCLUSION

- Restate the problem and motivation
- Summarize the eQual framework architecture
- Highlight key findings:
  * Multi-level bias detection effectiveness
  * Validated neutral rewriting with semantic preservation
  * Fairness-aware recommendation performance
  * Cross-domain generalization capability
  * Data augmentation success for implicit bias
- Emphasize contributions to NLP fairness and ICT applications
- Acknowledge limitations
- Outline future research directions
- Closing statement on inclusive recruitment

6. ACKNOWLEDGEMENTS

- Funding sources
- Data providers (PESO/DOLE)
- Annotators and participants

7. AUTHOR CONTRIBUTIONS (CRediT)

- Conceptualization: [Authors]
- Methodology: [Authors]
- Software: [Authors]
- Validation: [Authors]
- Formal Analysis: [Authors]
- Investigation: [Authors]

- Resources: [Authors]
- Data Curation: [Authors]
- Writing – Original Draft: [Authors]
- Writing – Review & Editing: [Authors]
- Visualization: [Authors]
- Supervision: [Authors]
- Project Administration: [Authors]
- Funding Acquisition: [Authors]

8. REFERENCES

- Minimum 25 references (15 recent journal articles)
- IEEE format with DOIs
- Sequential numbering
- All cited in text

9. AUTHOR BIOGRAPHIES

- Include links to Scholar, Scopus, Publons, ORCID
- Follow IJICT template

V. DETAILED IMPLEMENTATION SPECIFICATIONS (NEW SECTION)

Purpose: Address Reviewer B's reproducibility requirement

A. Model Configurations

Bias Detection Models:
├─── Rule-Based Lexicon:
│ ├─── Lexicon source: [Gaucher et al. 2011] + custom Philippine terms
│ ├─── Matching algorithm: Case-insensitive substring matching
│ └─── Context window: ±5 words
├─── TF-IDF Classifier:
│ ├─── Vectorizer: TfidfVectorizer (sklearn)
│ ├─── Max features: 5000
│ ├─── N-gram range: (1, 3)
│ ├─── Classifier: Logistic Regression
│ ├─── Regularization: L2, C=1.0
│ └─── Solver: 'lbfgs', max_iter=1000
└─── Transformer Model:
├─── Base model: RoBERTa-base (roberta-base)
├─── Fine-tuning:
│ ├─── Learning rate: 2e-5
│ ├─── Batch size: 16
│ ├─── Epochs: 5

```
    |   ├── Warmup steps: 500
    |   ├── Weight decay: 0.01
    |   ├── Optimizer: AdamW
    |   └── Scheduler: Linear with warmup
    ├── Max sequence length: 512 tokens
    ├── Dropout: 0.1
    └── Classification head: Linear layer (768 → 2)

Neutral Rewriting Models:
 ├── Rule-Based:
 |   ├── Substitution dictionary: 150+ gendered terms → neutral
 |   └── Context-aware rules: 25 patterns
 ├── Neural Model:
 |   ├── Base model: T5-base or BART-base
 |   ├── Fine-tuning:
 |   |   ├── Learning rate: 3e-5
 |   |   ├── Batch size: 8
 |   |   ├── Epochs: 10
 |   |   ├── Max input length: 512
 |   |   ├── Max output length: 512
 |   |   └── Beam search: width=5
 |   └── Training data: Original + augmented pairs
 └── LLM-Guided:
     ├── Model: GPT-4 (gpt-4-turbo)
     ├── Temperature: 0.7
     ├── Top-p: 0.9
     ├── Max tokens: 500
     └── Prompt template: [Detailed prompt in appendix]

Recommendation System:
 ├── Ranking algorithm: BM25 + Semantic similarity
 ├── Semantic model: Sentence-BERT (all-MiniLM-L6-v2)
 ├── Fairness constraint: Exposure fairness threshold=0.8
 └── Diversity parameter: λ=0.3 (relevance-diversity tradeoff)
```

B. Training Procedures

Training Pipeline:
1. Data Preprocessing:
   - Text cleaning: Remove URLs, special characters
   - Tokenization: RoBERTa tokenizer

- Label encoding: Binary (biased=1, neutral=0)

2. Data Splitting:
- Training: 70% (1,051 samples)
- Validation: 15% (225 samples)
- Test: 15% (225 samples)
- Stratified split to preserve class distribution

3. Training Loop:
- Early stopping: Patience=3, monitor validation F1
- Gradient clipping: Max norm=1.0
- Mixed precision training: FP16
- Checkpointing: Save best model by validation F1

4. Hyperparameter Tuning:
- Method: Grid search on validation set
- Parameters: Learning rate, batch size, epochs
- Evaluation: 5-fold cross-validation

5. Data Augmentation (for implicit bias):
- Apply augmentation after initial training
- Curriculum learning: Original → Augmented (50/50) → Augmented (100%)
- Re-train with augmented data for 3 additional epochs

C. Hardware and Software Specifications

Hardware:
├── GPU: NVIDIA Tesla V100 (32GB) or equivalent
├── CPU: Intel Xeon (16 cores)
├── RAM: 64GB
└── Storage: 500GB SSD

Software:
├── Operating System: Ubuntu 20.04 LTS
├── Python: 3.8.10
├── PyTorch: 1.12.0
├── Transformers (Hugging Face): 4.20.0
├── Scikit-learn: 1.1.1
├── NLTK: 3.7
├── Sentence-Transformers: 2.2.0
├── OpenAI API: 0.27.0 (for GPT-4)
└── Other dependencies: pandas, numpy, matplotlib, seaborn

Training Time:
├── RoBERTa fine-tuning: ~2 hours (5 epochs)

├─── T5/BART fine-tuning: ~4 hours (10 epochs)
├─── Data augmentation: ~6 hours (500 samples via GPT-4)
└─── Total pipeline: ~12-15 hours

D. Code and Data Availability

Reproducibility Package:
├─── Code Repository: [GitHub URL]
│ ├─── README.md (setup instructions)
│ ├─── requirements.txt (dependencies)
│ ├─── data/ (sample data, full data upon request)
│ ├─── models/ (pre-trained model checkpoints)
│ ├─── scripts/
│ │ ├─── train_classifier.py
│ │ ├─── train_rewriter.py
│ │ ├─── evaluate.py
│ │ ├─── augment_data.py
│ │ └─── run_pipeline.py
│ └─── notebooks/ (Jupyter notebooks for analysis)
├─── Dataset:
│ ├─── PESO job ads: Available upon request (privacy considerations)
│ ├─── Gold-400 annotations: Included in repository
│ └─── Augmented data: Included in repository
└─── Pre-trained Models:
├─── RoBERTa bias classifier: [Hugging Face URL]
├─── T5 rewriter: [Hugging Face URL]
└─── Sentence-BERT embeddings: [Hugging Face URL]

VI. HUMAN EVALUATION PROTOCOL (MANDATORY)

Purpose: Address Reviewer B's requirement for completed human evaluation

A. Evaluation Design

Human Evaluation Study:
├─── Objective: Assess neutral rewriting quality
├─── Sample Size: 100 job ad segments (randomly selected from test set)
├─── Annotators:
│ ├─── N=3 per sample (total 9 annotators)
│ ├─── Background: HR professionals (3), linguists (3), domain experts (3)
│ ├─── Training: 2-hour session on evaluation criteria

| └── Compensation: [Amount] per hour
├── Evaluation Interface: Custom web-based annotation tool
└── Timeline: 2 weeks

B. Evaluation Criteria

Rating Scales (5-point Likert):
1. Neutrality (1=Highly biased, 5=Completely neutral)
- Does the rewritten text remove gender bias?
- Are there any remaining gendered cues?

2. Semantic Preservation (1=Completely different, 5=Identical meaning)
- Does the rewritten text preserve the original job requirements?
- Are key qualifications and responsibilities maintained?

3. Fluency/Clarity (1=Incomprehensible, 5=Perfectly fluent)
- Is the rewritten text grammatically correct?
- Is it clear and easy to understand?

4. Acceptability (Binary: Acceptable / Not Acceptable)
- Would you accept this rewrite for a real job posting?

Additional Questions:
- Which version would attract more diverse applicants? (Original / Rewritten / No difference)
- Any specific concerns or suggestions? (Open-ended)

C. Inter-Rater Reliability

Reliability Metrics:
├── Krippendorff's Alpha: Target ≥ 0.70
├── Intraclass Correlation Coefficient (ICC): Target ≥ 0.75
├── Percentage Agreement: Report for binary acceptability
└── Disagreement Resolution: Majority vote or expert adjudication

D. Analysis Plan

Statistical Analysis:
├── Descriptive Statistics:
| ├── Mean, median, SD for each criterion
| ├── Distribution plots (histograms, box plots)
| └── Correlation between criteria
├── Comparative Analysis:
| ├── Compare rule-based vs. neural vs. LLM-guided rewriting
| ├── Paired t-tests or Wilcoxon signed-rank tests
| └── Effect sizes (Cohen's d)
├── Qualitative Analysis:

```
│   ├──── Thematic analysis of open-ended feedback
│   ├──── Common failure modes
│   └──── Improvement suggestions
└──── Reporting:
    ├──── Table: Mean scores with 95% CI for each criterion
    ├──── Figure: Comparison across rewriting methods
    └──── Discussion: Strengths, weaknesses, implications
```

VII. CROSS-DOMAIN VALIDATION PROTOCOL (NEW)

Purpose: Address Reviewer B's requirement for external validation

A. External Datasets

```
Dataset Selection:
├──── Dataset 1: Australian Job Ads [2]
│   ├──── Source: Mao et al. 2023 study
│   ├──── Size: ~12M job ads (sample 1,000 for testing)
│   ├──── Domain: General employment (ICT, manufacturing, services)
│   └──── Purpose: Test generalization to international context
├──── Dataset 2: STEM Academic Job Postings [3]
│   ├──── Source: Dikshit et al. 2024 study
│   ├──── Size: ~500 academic job ads
│   ├──── Domain: STEM faculty positions
│   └──── Purpose: Test generalization to academic sector
└──── Dataset 3: Private Sector Job Ads (Philippine)
    ├──── Source: Collect from JobStreet, LinkedIn (with permission)
    ├──── Size: 500 job ads
    ├──── Domain: Private companies (non-PESO)
    └──── Purpose: Test generalization within Philippine context
```

B. Evaluation Protocol

```
Cross-Domain Testing:
├──── Zero-Shot Transfer:
│   ├──── Apply eQual models directly (no retraining)
│   ├──── Measure performance drop from source domain
│   └──── Identify failure modes
├──── Few-Shot Adaptation:
│   ├──── Fine-tune with 10, 50, 100 samples from target domain
│   ├──── Measure adaptation effectiveness
```

```
│   └── Compare with zero-shot performance
├── Metrics:
│   ├── Classification: Accuracy, F1, Precision, Recall
│   ├── Generalization Gap: |F1_source - F1_target|
│   ├── Domain Similarity: A-distance, MMD
│   └── Robustness: Performance variance across domains
└── Analysis:
    ├── Error analysis: Common misclassifications
    ├── Domain-specific challenges
    └── Recommendations for domain adaptation
```

C. Overfitting Detection

```
Overfitting Analysis:
├── Learning Curves:
│   ├── Plot training vs. validation loss/accuracy
│   ├── Identify overfitting indicators (divergence)
│   └── Determine optimal stopping point
├── Performance Gap:
│   ├── Train-Test Gap: |Acc_train - Acc_test|
│   ├── Source-Target Gap: |Acc_source - Acc_target|
│   └── Threshold: Gap > 10% indicates potential overfitting
├── Regularization Effectiveness:
│   ├── Compare models with/without dropout, weight decay
│   ├── Ablation study on regularization parameters
│   └── Report optimal configuration
└── Mitigation Strategies (if overfitting detected):
    ├── Increase dropout rate
    ├── Apply data augmentation
    ├── Reduce model complexity
    └── Early stopping with stricter patience
```

VIII. FIGURES AND TABLES ENHANCEMENT

A. New Figures to Add

```
Figure 1: eQual Framework Architecture (Flowchart)
├── Components: Data Collection → Annotation → Bias Detection →
│               Span Localization → Rewriting → Recommendation →
│               Validation → Deployment
├── Highlight: Modular design, extensibility points
```

└── Format: Professional flowchart with clear labels

Figure 2: Bias Detection Pipeline
├── Multi-level architecture: Explicit, Implicit, Intersectional
├── Model comparison: Rule-based, TF-IDF, Transformer
└── Output: Bias scores, spans, categories

Figure 3: Data Augmentation Process
├── LLM-based generation workflow
├── Quality control steps
└── Before/after class distribution

Figure 4: Human Evaluation Results
├── Box plots for Neutrality, Semantic Preservation, Fluency
├── Comparison across rewriting methods
└── Inter-rater reliability visualization

Figure 5: Cross-Domain Performance
├── Bar chart: Performance across source and target domains
├── Generalization gap visualization
└── Domain adaptation effectiveness

Figure 6: Learning Curves
├── Training vs. validation loss/accuracy
├── Overfitting detection
└── Optimal stopping point

Figure 7: Confusion Matrices
├── Explicit bias detection
├── Implicit bias detection
└── Cross-domain performance

Figure 8: Fairness Metrics Comparison
├── Demographic Parity Difference
├── Equalized Odds Difference
└── Counterfactual fairness scores

B. Tables Enhancement

Table 1: Dataset Statistics (Enhanced)
├── Add: Implicit vs. explicit bias distribution
├── Add: Augmented data statistics
└── Add: Cross-domain dataset comparison

Table 2: Bias Detection Performance (Enhanced)
├─── Add: Per-class precision/recall
├─── Add: Fairness metrics (DPD, EOD)
└─── Add: Comparison with baselines

Table 3: Rewriting Evaluation (NEW)
├─── Automatic metrics: BLEU, ROUGE, BERTScore
├─── Semantic preservation: Sentence-BERT similarity
├─── Bias reduction: Before/after scores
└─── Human evaluation: Mean scores with 95% CI

Table 4: Cross-Domain Validation Results (NEW)
├─── Performance on each external dataset
├─── Generalization gap
└─── Domain adaptation effectiveness

Table 5: Ablation Study (NEW)
├─── Impact of each component (detection, rewriting, recommendation)
├─── Impact of data augmentation
└─── Impact of fairness constraints

Table 6: Comparison with State-of-the-Art (NEW)
├─── eQual vs. [2], [4], [6]
├─── Metrics: Accuracy, F1, Fairness scores
└─── Advantages/disadvantages

Table 7: Implementation Specifications (NEW)
├─── Model configurations
├─── Hyperparameters
├─── Training time
└─── Hardware requirements

Table 8: Human Evaluation Inter-Rater Reliability (NEW)
├─── Krippendorff's Alpha
├─── ICC
└─── Percentage agreement

C. Table Integration in Text

Requirement (Reviewer A & C): Introduce each table in text before display

Example:

"Table 1 presents the comprehensive statistics of our PESO job advertisement dataset, including the distribution of explicit and implicit bias examples before and after data augmentation. As shown in Table 1, the original dataset contained 1,501 job ads, with [X]% exhibiting explicit bias and only [Y]% showing implicit bias. After applying our LLM-based augmentation strategy, the implicit bias examples increased to [Z]%, significantly improving class balance."

[TABLE 1 APPEARS HERE]

"The improved class distribution enabled more robust training of our implicit bias detection model, as discussed in Section 4.2.1."

IX. CITATION MANAGEMENT

A. Current Issues

- Inconsistent numbering sequence
- Some references not cited in text (11 , 12 , 13 , 14 , 16 , 18)
- Need minimum 25 references (15 recent journal articles)

B. Citation Strategy

Citation Plan:
├─── Total References: 30-35 (exceeds minimum 25)
├─── Recent Journal Articles: 18-20 (exceeds minimum 15)
├─── Conference Papers: 8-10
├─── Books/Reports: 2-3
└─── Online Resources: 1-2 (PESO/DOLE guidelines)

Citation Distribution by Section:
├─── Introduction: 8-10 citations
│ ├─── Gender bias impact studies [1], [2], [3]
│ ├─── Philippine employment context
│ └─── NLP fairness overview
├─── Related Work: 12-15 citations
│ ├─── Bias detection methods [1], [4], [5], [6], [7], [8]
│ ├─── Rewriting approaches [20], [21], [27], [28], [29]
│ ├─── Fairness metrics [Measuring Fairness 2022]
│ └─── Recommendation systems [17]
├─── Methodology: 5-8 citations
│ ├─── Model architectures [RoBERTa, BERT papers]
│ ├─── Evaluation metrics [10], [15]
│ └─── Data augmentation [23], [AugGPT 2023]

├─── Results & Discussion: 8-12 citations
│ ├─── Comparison with [2], [4], [6]
│ ├─── Fairness analysis [Fairness metrics papers]
│ └─── Limitations discussion
└─── Conclusion: 2-3 citations
└─── Future directions

C. IEEE Format Compliance

Reference Format (IEEE):
[1] A. Author, B. Author, and C. Author, "Title of paper," Journal Name, vol. X, no. Y, pp. ZZ-ZZ, Month Year, doi: XX.XXXX/XXXXXX.

[2] A. Author and B. Author, "Title of paper," in Proc. Conference Name, City, Country, Year, pp. ZZ-ZZ, doi: XX.XXXX/XXXXXX.

[3] A. Author, Title of Book, Edition. City, Country: Publisher, Year.

Tools:
├─── Reference Manager: Mendeley, Zotero, or EndNote
├─── Style: IEEE
├─── DOI: Mandatory for all journal articles and conference papers
└─── Verification: Check all references have DOIs where available

D. Citation Verification Checklist

Pre-Submission Checklist:
☐ All references cited in text (no orphan references)
☐ All in-text citations have corresponding references
☐ Sequential numbering (first appearance order)
☐ IEEE format compliance
☐ DOIs included for all eligible references
☐ Minimum 25 references (target: 30-35)
☐ Minimum 15 recent journal articles (target: 18-20)
☐ No duplicate references
☐ Author names, titles, venues verified
☐ Page numbers included where applicable

X. WRITING QUALITY IMPROVEMENTS

A. Clarity and Conciseness

Current Issue (Reviewer A): "Too many technical terms and secondary references"

Improvement Strategy:

Writing Guidelines:

- Simplify Technical Jargon:
  - Define acronyms on first use
  - Use plain language where possible
  - Add glossary if needed
- Reduce Secondary References:
  - Cite primary sources directly
  - Limit "as cited in" references
  - Verify original sources
- Improve Flow:
  - Use transition sentences between sections
  - Maintain consistent terminology
  - Avoid repetition
- Enhance Readability:
  - Vary sentence length
  - Use active voice
  - Break long paragraphs (max 150 words)

B. Discussion Structure (6-Step Approach)

Associate Editor Requirement: Follow traditional 6-step discussion structure

Discussion Template:

1. Introduction – Mention gaps in previous research
   "This study investigated gender bias detection and mitigation in Philippine public employment job advertisements. While earlier studies have explored bias detection in Western contexts [2], [4], they have not explicitly addressed the unique linguistic and cultural challenges in Philippine employment platforms, nor have they provided end-to-end frameworks with validated rewriting and fairness-aware recommendation."

2. Summarizing key findings – Let your data speak
   "We found that transformer-based models (RoBERTa) significantly outperform rule-based and TF-IDF approaches in detecting both explicit (F1=X.XX) and implicit bias (F1=Y.YY). The proposed data augmentation strategy increased implicit bias examples by Z%, leading to a W% improvement in minority class recall. Human evaluation confirmed that our neutral rewriting module achieves high neutrality (M=X.X/5.0) while preserving semantic meaning (M=Y.Y/5.0)."

3. Interpreting results – Compare with other papers
   "Our results align with [2], which reported similar performance gains using RoBERTa for Australian job ads. However, our framework extends beyond detection to include validated rewriting and fairness-aware recommendation, addressing a gap identified in [4]. Unlike [6], which

focused solely on explicit bias, our multi-level approach captures subtle implicit biases through agentic-communal classification."

4. Addressing limitations – Their potential impact on the results
"This study explored a comprehensive dataset of 1,501 PESO job advertisements with augmented implicit bias examples. However, further and in-depth studies may be needed to confirm generalization to private sector job ads and other languages (e.g., Filipino, Tagalog), especially regarding cross-cultural bias patterns. The limited sample size for human evaluation (N=100) may affect the robustness of rewriting quality assessments."

5. Implications for future research – How to explore further
"Our study demonstrates that LLM-based data augmentation and multi-level bias detection are effective for addressing class imbalance and improving fairness. Future studies may explore real-time bias detection in live recruitment platforms, multi-modal bias analysis (text + images), and longitudinal impact studies on applicant diversity and hiring outcomes."

6. Conclusion – Summarize content
"Recent observations suggest that gender-biased language in job advertisements significantly impacts applicant diversity. Our findings provide conclusive evidence that the eQual framework can effectively detect, rewrite, and recommend neutral alternatives, contributing to more inclusive recruitment practices in Philippine public employment services."

XI. TIMELINE AND MILESTONES

Phase 1: Literature Integration (2 weeks)

- [ ] Review 696 papers from 5 literature searches
- [ ] Extract key methodologies, metrics, and insights
- [ ] Identify gaps and positioning for eQual
- [ ] Draft Related Work section

Phase 2: Data Augmentation (3 weeks)

- [ ] Design LLM-based augmentation prompts
- [ ] Generate 500+ implicit bias examples
- [ ] Quality control and validation
- [ ] Re-train models with augmented data
- [ ] Document results

Phase 3: Human Evaluation (2 weeks)

- [ ] Recruit and train annotators (N=9)
- [ ] Develop annotation interface
- [ ] Conduct evaluation (100 samples)

- [ ] Analyze results and compute reliability
- [ ] Write up findings

Phase 4: Cross-Domain Validation (3 weeks)

- [ ] Acquire external datasets (Australian, STEM, Private)
- [ ] Prepare data for testing
- [ ] Run zero-shot and few-shot experiments
- [ ] Analyze generalization performance
- [ ] Document overfitting analysis

Phase 5: Implementation Documentation (1 week)

- [ ] Document model configurations
- [ ] Document hyperparameters and training procedures
- [ ] Document hardware/software specifications
- [ ] Prepare code repository
- [ ] Create reproducibility package

Phase 6: Manuscript Revision (4 weeks)

- [ ] Restructure paper (IMRADC format)
- [ ] Add methodology flowchart
- [ ] Integrate new results (augmentation, human eval, cross-domain)
- [ ] Enhance discussion (6-step structure)
- [ ] Add/revise figures and tables
- [ ] Introduce tables in text
- [ ] Fix citations (sequential, IEEE format, DOIs)
- [ ] Simplify language and improve clarity
- [ ] Add implementation details section

Phase 7: Quality Assurance (1 week)

- [ ] Internal review by co-authors
- [ ] Check all reviewer requirements addressed
- [ ] Verify citation compliance
- [ ] Proofread for grammar and style
- [ ] Format according to IJICT template
- [ ] Prepare author biographies with links

Phase 8: Submission (1 week)

- [ ] Final formatting check
- [ ] Prepare cover letter addressing reviewer comments
- [ ] Upload to journal system
- [ ] Submit supplementary materials (code, data)

Total Timeline: 17 weeks (~4 months)

XII. RISK MITIGATION

Risk 1: Human Evaluation Delays

Mitigation:

- Start annotator recruitment early
- Have backup annotators
- Use online platform for remote evaluation
- Provide clear instructions and training

Risk 2: External Dataset Access

Mitigation:

- Contact authors of 2 , 3 for dataset access
- Prepare alternative datasets (e.g., Kaggle, GitHub)
- Use web scraping (with permission) for private sector ads
- Document data acquisition process

Risk 3: Computational Resources

Mitigation:

- Use cloud computing (Google Colab Pro, AWS)
- Optimize code for efficiency
- Use smaller models if needed (DistilBERT, DistilRoBERTa)
- Parallelize experiments

Risk 4: Timeline Overruns

Mitigation:

- Prioritize critical enhancements (human eval, cross-domain)
- Defer optional enhancements if needed
- Allocate buffer time (2 weeks)
- Regular progress monitoring

XIII. SUCCESS CRITERIA

Minimum Requirements (Must-Have)

- ☐ IMRADC structure compliance
- ☐ Methodology flowchart added
- ☐ Human evaluation completed and reported
- ☐ Cross-domain validation conducted
- ☐ Implementation details documented
- ☐ All tables introduced in text
- ☐ Citations fixed (sequential, IEEE, DOIs)
- ☐ Minimum 25 references (15 recent journals)

- [ ] Author biographies with links
- [ ] Discussion follows 6-step structure

Enhanced Requirements (Should-Have)

- [ ] Data augmentation for implicit bias
- [ ] Fairness metrics (DPD, EOD) reported
- [ ] Comparison with state-of-the-art
- [ ] Ablation studies
- [ ] Error analysis
- [ ] Qualitative analysis of failures
- [ ] Related Work section added

Aspirational Requirements (Nice-to-Have)

- [ ] Code and data publicly available
- [ ] Pre-trained models on Hugging Face
- [ ] Interactive demo
- [ ] Video presentation
- [ ] Supplementary materials (appendices)

XIV. CONCLUSION

This enhancement plan provides a comprehensive roadmap to transform the eQual paper from a system description to a rigorous framework paper that addresses all reviewer concerns. The plan integrates insights from 696 related studies across five literature domains, ensuring the paper is well-positioned within the NLP fairness landscape.

Key Enhancements:

1. Framework Positioning: Emphasize modularity, extensibility, and adaptability
2. Validation Rigor: Human evaluation, cross-domain testing, fairness metrics
3. Data Augmentation: Address implicit bias scarcity with LLM-based generation
4. Reproducibility: Detailed implementation specifications and code availability
5. Discussion Depth: 6-step structure, comparison with SOTA, limitations, implications

Expected Outcomes:

- Addresses all critical reviewer comments
- Strengthens methodological rigor
- Enhances practical relevance for ICT applications
- Positions eQual as a leading framework for gender-bias detection and mitigation in job advertisements
- Increases likelihood of acceptance in high-quality journal

Next Steps:

1. Review this enhancement plan with co-authors

2. Prioritize enhancements based on timeline and resources
3. Begin implementation (start with human evaluation and cross-domain validation)
4. Regular progress monitoring and adjustment
5. Iterative manuscript revision

APPENDICES

Appendix A: Recommended Literature by Category

Bias Detection (Top 10):

1. Context-Aware Discrimination Detection 1
2. Developing Large-Scale LM for Australian Job Ads 2
3. Investigating Gender Bias in STEM Job Ads 3
4. Balancing Gender Bias in Job Ads 4
5. Recognizing and Explaining Bias 5
6. UnMASKed: Quantifying Gender Biases 6
7. Transcending the "Male Code" 7
8. Measuring Bias in Contextualized Representations 8
9. Unveiling Gender Biases in Recruitment 9
10. NBIAS Framework 13

Neutral Rewriting (Top 10):

1. Detect and Perturb 20
2. Agency and Power Counterfactuals 21
3. Debiasing Job Ads by Replacing Masculine Language 22
4. NeuTral Rewriter 27
5. Syntax-guided Paraphrases 28
6. PowerTransformer 29
7. Gender-Neutral Rewriting in Italian 24
8. GenWriter 25
9. Exploiting Biased Models to De-bias Text 26
10. They, Them, Theirs 30

Validation & Fairness (Top 10):

1. Measuring Fairness via Prediction Sensitivity
2. NBIAS Framework 13
3. Assessing and Mitigating Social Bias [Tokpo 2024]
4. Fairness Evaluation in Text Classification [Ashktorab 2023]
5. Quantifying Social Biases in NLP [Czarnowska 2021]
6. Bias and Fairness in LLMs Survey [Gallegos 2024]
7. fairlib Framework [Han 2022]
8. On Intrinsic and Extrinsic Fairness Metrics
9. Bipol: Multi-Axes Bias Evaluation [Alkhaled 2023]
10. AutoTestForge [Xing 2025]

Cross-Domain & Generalization (Top 10):

1. Bridging Topic, Domain, Language Shifts [Waldis 2023]
2. Prompt-based Distribution Alignment [Chen 2022]
3. Training and Domain Adaptation [Glavaš 2021]
4. SEDGE: Specialized Ensemble Learning [Li 2022]
5. Improved Generalization of Arabic Classifiers [Khaddaj 2019]
6. Learning Semantic Topics [Fu 2018]
7. Domain Adaptation for NLP [various]
8. Transfer Learning for Few-Shot [various]
9. Meta-Learning for Domain Generalization [various]
10. Adversarial Domain Adaptation [various]

Data Augmentation & Class Imbalance (Top 10):

1. LLM-Guided Synthetic Augmentation 23
2. AugGPT
3. Employing Synthetic Data for ABSC [Ganganwar 2023]
4. Solving Data Imbalance with Contrastive Samples [Chen 2023]
5. Few-Shot Intent Classification [Kumar 2019]
6. Guiding Generative LMs for Data Augmentation [Edwards 2021]
7. PromptMix [Sahu 2023]
8. Large Model-Based Data Augmentation [Zhang 2024]
9. Fine-tuned Generative LLM Oversampling [Cloutier 2023]
10. Survey on Class Imbalance in Deep NLP [Henning 2023]

Appendix B: Prompt Templates for Data Augmentation

Prompt 1: Implicit Bias Generation

You are an expert in gender bias in job advertisements. Generate a job advertisement segment that contains IMPLICIT gender bias (not explicit gender requirements). The bias should be subtle and embedded in the language used to describe the role, qualifications, or work environment.

Context: [Job title, industry, level]
Bias Type: [Agentic/Communal]
Target Gender: [Male-coded/Female-coded]

Generate a 2-3 sentence job ad segment that:
1. Does NOT explicitly mention gender (no "male", "female", "he", "she")
2. Uses gender-coded language (agentic traits for male-coded, communal traits for female-coded)
3. Sounds natural and realistic for Philippine job advertisements
4. Is suitable for PESO/DOLE public employment platforms

Output format:
Biased Segment: [Generated text]
Bias Explanation: [Why this is implicitly biased]
Neutral Alternative: [Gender-neutral rewrite]

Prompt 2: Neutral Rewriting

You are an expert in inclusive language for job advertisements. Rewrite the following job ad segment to remove gender bias while preserving the original meaning and requirements.

Original Text: [Biased job ad segment]

Guidelines:
1. Remove all gendered language (explicit and implicit)
2. Replace gender-coded traits with neutral alternatives
3. Maintain the same job requirements and qualifications
4. Keep the tone professional and appropriate for Philippine employment
5. Ensure the rewrite is clear, fluent, and natural

Output format:
Neutral Rewrite: [Generated text]
Changes Made: [List of specific changes]
Preserved Elements: [What was kept from original]

Appendix C: Human Evaluation Interface Mockup

[SCREENSHOT/WIREFRAME OF ANNOTATION TOOL]

Components:
├─── Instructions Panel (left)
├─── Original Job Ad Segment (top center)
├─── Rewritten Job Ad Segment (bottom center)
├─── Rating Scales (right):
│ ├─── Neutrality: ○ 1 ○ 2 ○ 3 ○ 4 ○ 5
│ ├─── Semantic Preservation: ○ 1 ○ 2 ○ 3 ○ 4 ○ 5
│ ├─── Fluency: ○ 1 ○ 2 ○ 3 ○ 4 ○ 5
│ └─── Acceptability: ○ Acceptable ○ Not Acceptable
├─── Comparison Question:
│ "Which version would attract more diverse applicants?"
│ ○ Original ○ Rewritten ○ No difference
├─── Comments Box (bottom)
└─── Navigation: [Previous] [Next] [Save] [Submit]

Appendix D: Cross-Domain Dataset Acquisition Plan

Dataset 1: Australian Job Ads

- Contact: Ruochen Mao (rmao@student.unimelb.edu.au)
- Request: Sample of 1,000 job ads with bias annotations
- Backup: Use publicly available job ad datasets from Kaggle

Dataset 2: STEM Academic Job Postings

- Contact: Malika Dikshit (mdikshit@hbku.edu.qa)
- Request: Dataset from "Investigating Gender Bias in STEM Job Ads" paper
- Backup: Collect from HigherEdJobs.com, AcademicKeys.com (with permission)

Dataset 3: Philippine Private Sector

- Source: JobStreet Philippines, LinkedIn Philippines
- Method: Web scraping with robots.txt compliance
- Sample: 500 job ads across industries
- Annotation: Manual annotation by team (explicit/implicit bias labels)

END OF ENHANCEMENT PLAN

This plan serves as a comprehensive reference for revising the eQual framework paper. No enhancements should be made at this point; this document establishes the roadmap for future revisions based on reviewer feedback and literature insights.

References (29)

[1]"Context-Aware Discrimination Detection in Job Vacancies using Computational Language Models," Feb. 2022, doi: 10.48550/arxiv.2202.03907.
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